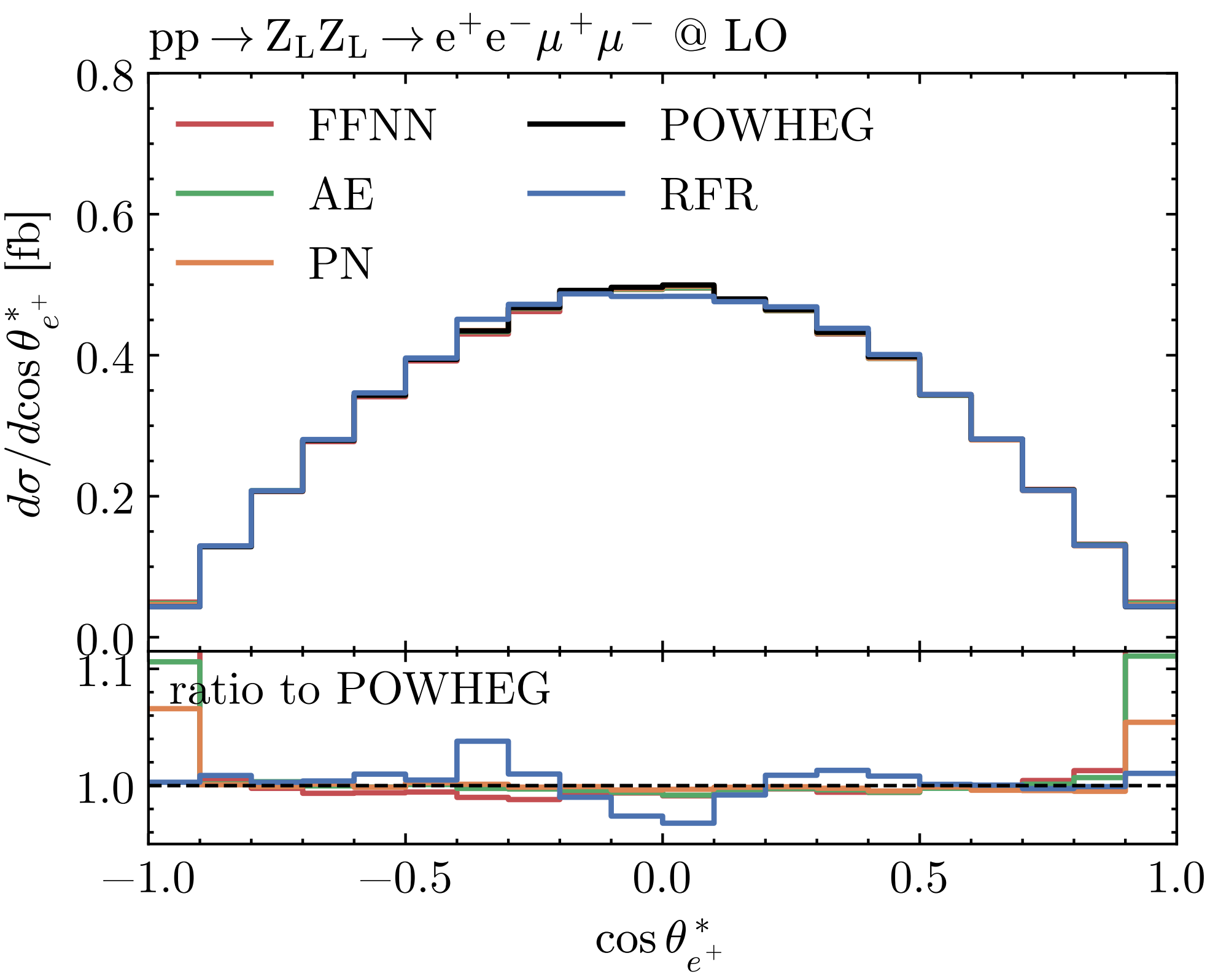
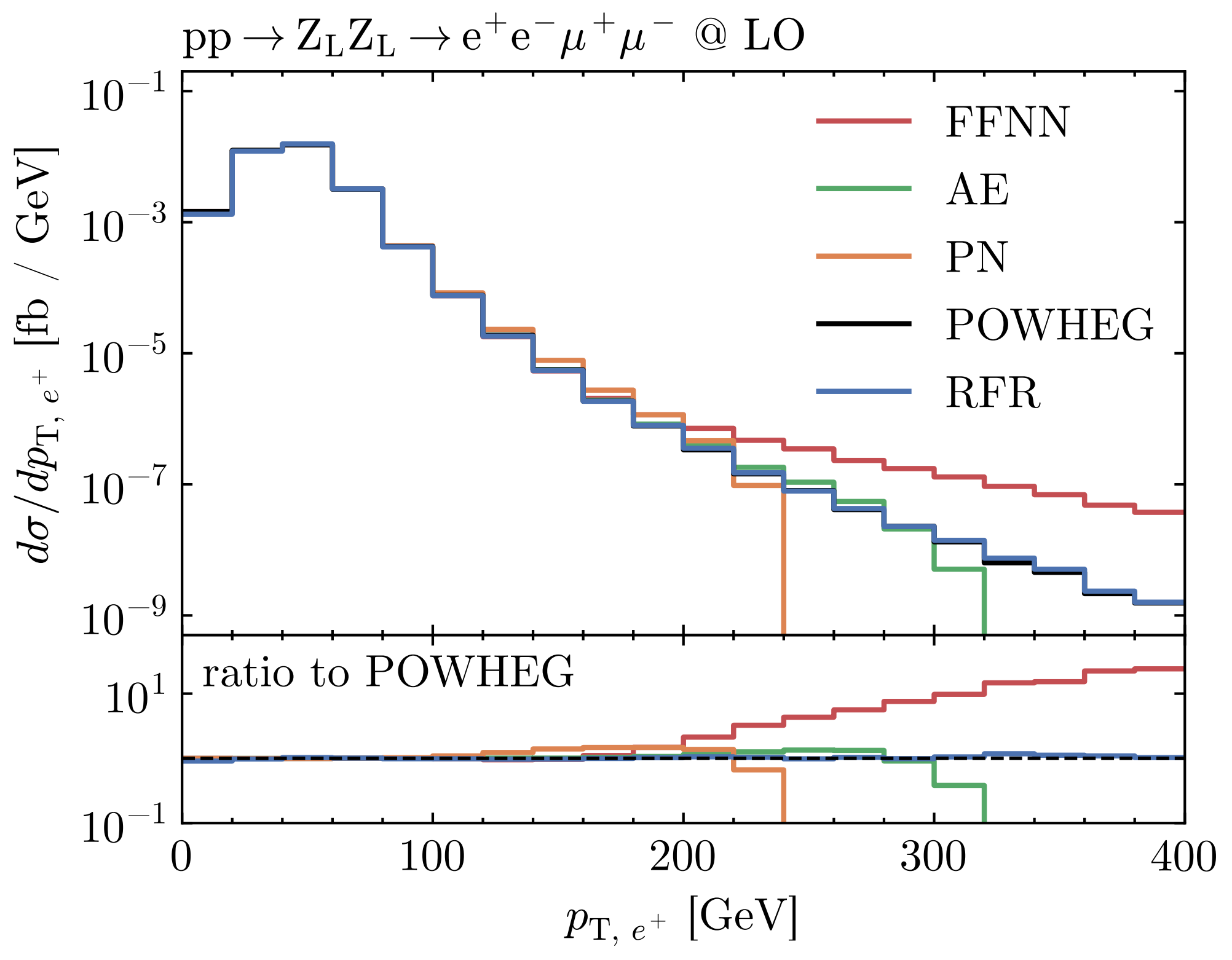
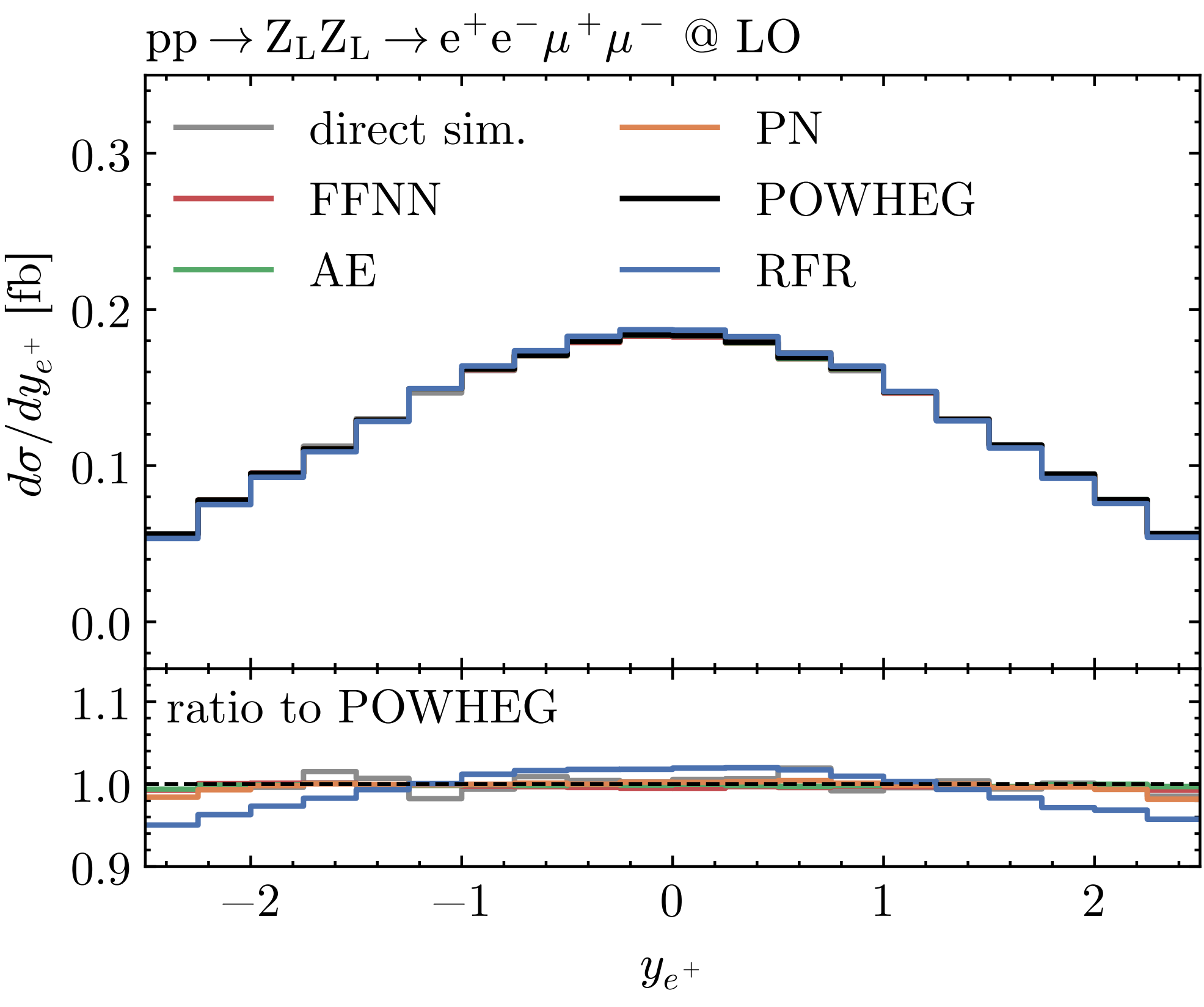
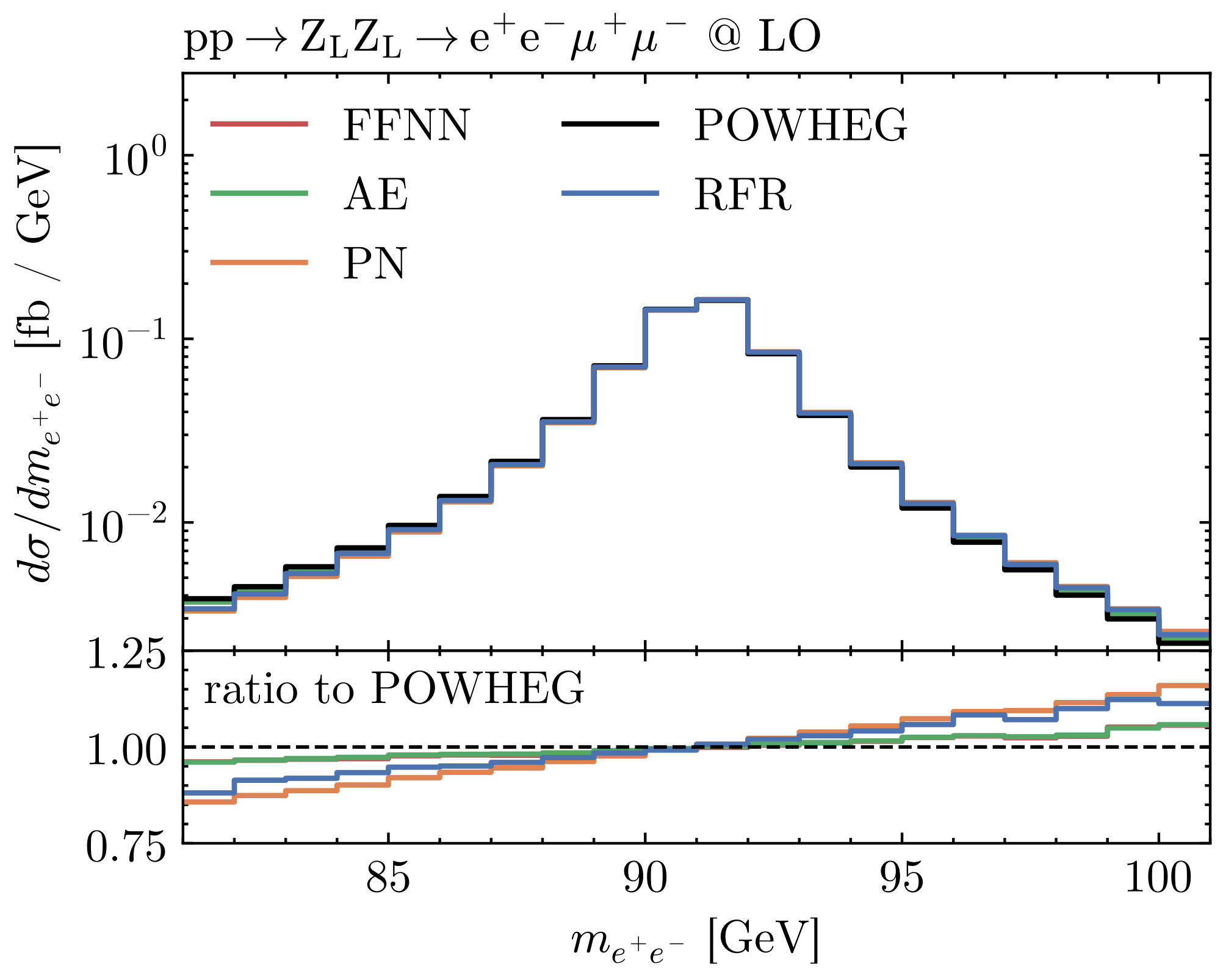
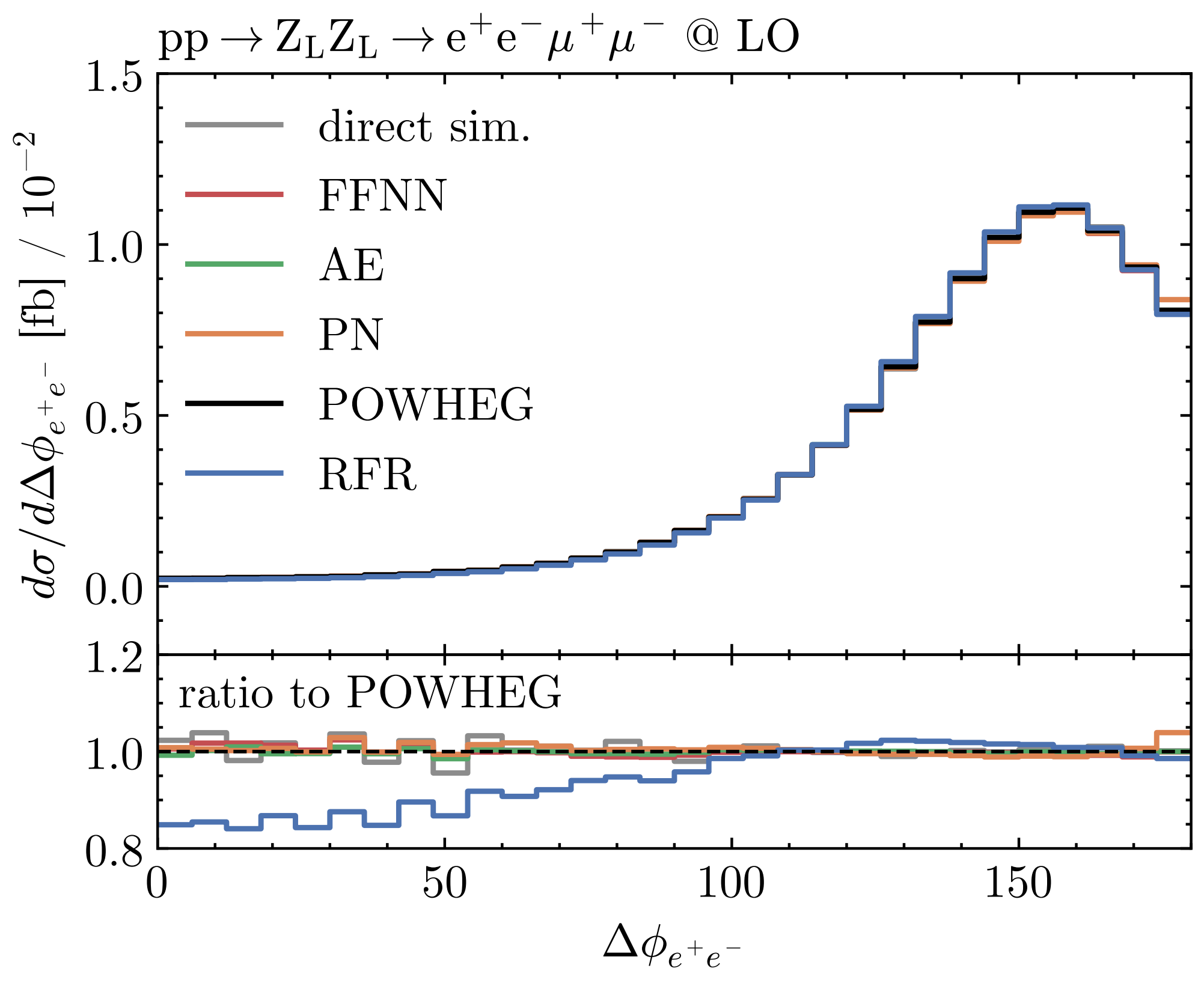
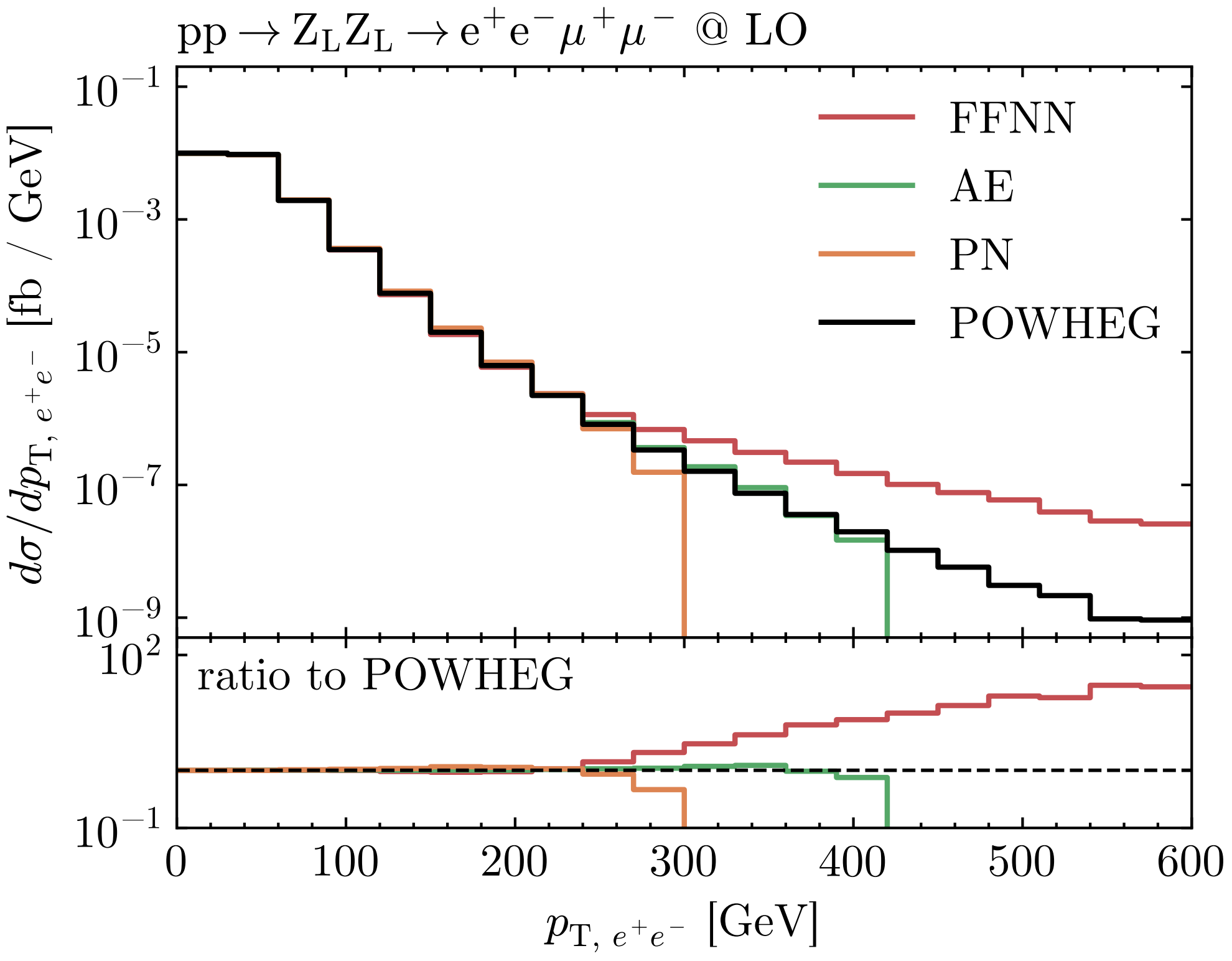

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
FFNN
AE
PN
POWHEG
RFR
$d\sigma / d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

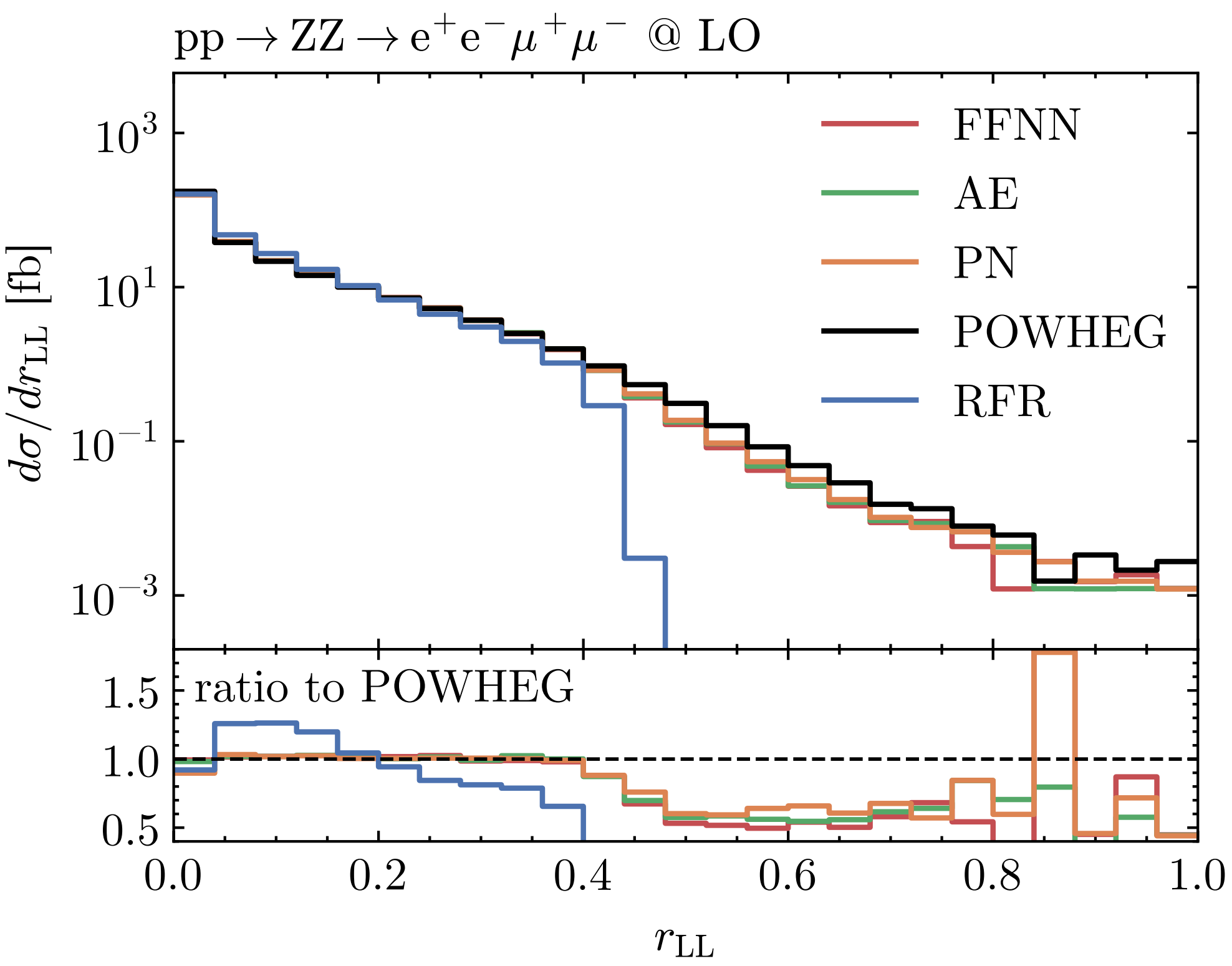

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dp_{\mathrm{T},\,e^+}$ [fb / GeV]
ratio to POWHEG
$p_{\mathrm{T},\,e^+}$ [GeV]

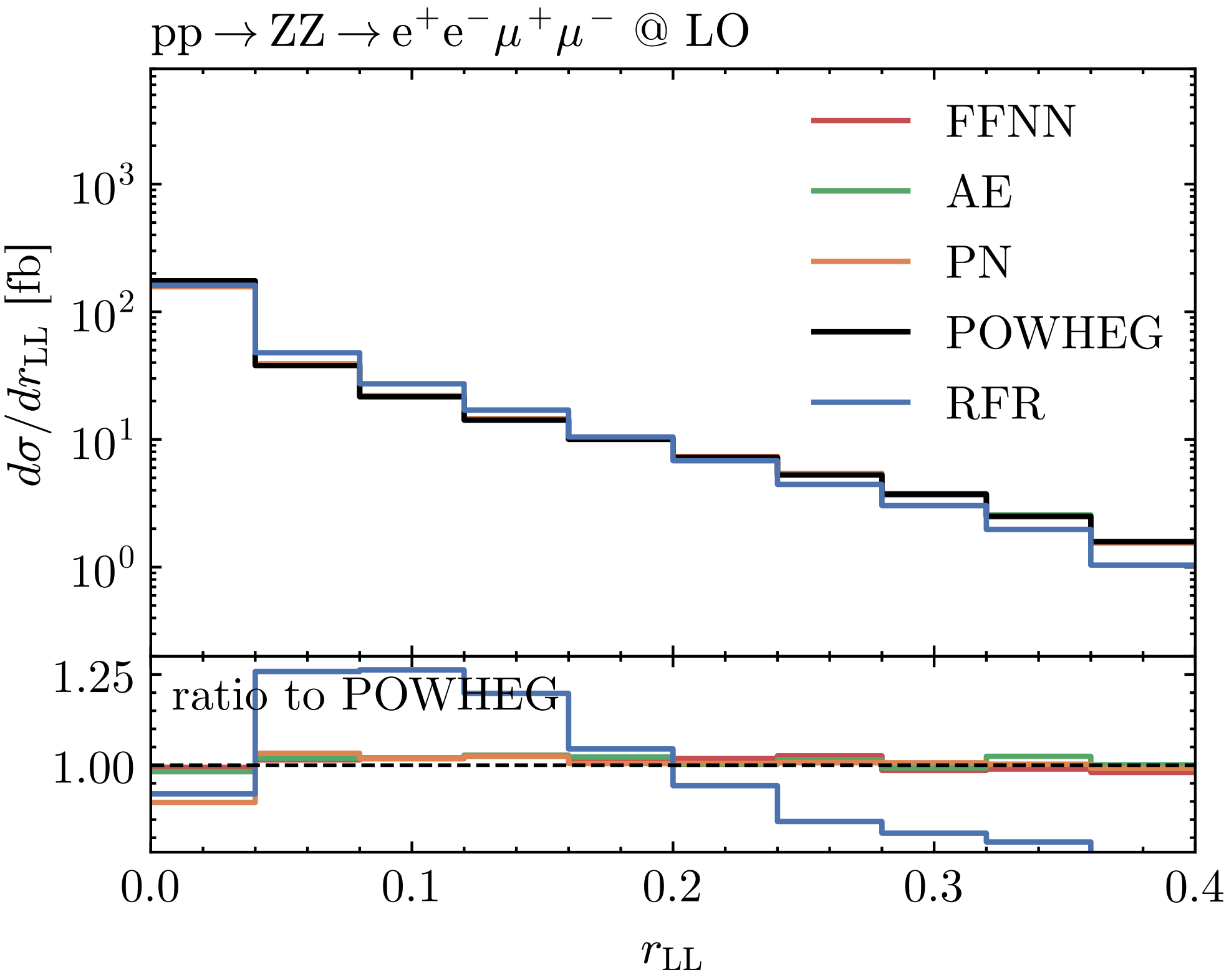

pp → $Z_L Z_L$ → $e^+e^-\mu^+\mu^-$ @ LO
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dy_{e^+}$ [fb]
0.3
0.2
0.1
0.0
ratio to POWHEG
1.1
1.0
0.9
−2
−1
0
1
2
$y_{e^+}$


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻ @ LO
FFNN
AE
PN
POWHEG
RFR
10^0
10^−1
10^−2
dσ/dm_{e⁺e⁻} [fb / GeV]
1.25
1.00
0.75
ratio to POWHEG
85
90
95
100
m_{e⁺e⁻} [GeV]


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma / d\Delta\phi_{e^+e^-}$ [fb] / $10^{-2}$
1.5
1.0
0.5
0.0
ratio to POWHEG
1.2
1.0
0.8
0
50
100
150
$\Delta\phi_{e^+e^-}$


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
FFNN
AE
PN
POWHEG
$d\sigma/dp_{\mathrm{T},\,e^+e^-}$ [fb / GeV]
ratio to POWHEG
$p_{\mathrm{T},\,e^+e^-}$ [GeV]
$10^{-1}$
$10^{-3}$
$10^{-5}$
$10^{-7}$
$10^{-9}$
$10^{2}$
$10^{-1}$
0
100
200
300
400
500
600


pp → ZZ → e+e−μ+μ− @ LO
FFNN
AE
PN
POWHEG
RFR
dσ/dr_LL [fb]
ratio to POWHEG
r_LL


$\mathrm{pp} \to \mathrm{ZZ} \to \mathrm{e}^+\mathrm{e}^-\mu^+\mu^-$ @ LO
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dr_{\mathrm{LL}}$ [fb]
$10^3$
$10^2$
$10^1$
$10^0$
ratio to POWHEG
1.25
1.00
0.0
0.1
0.2
0.3
0.4
$r_{\mathrm{LL}}$